

AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Faculty of Science and Technology
Department of Computer Science

CSC 2210 - Object Oriented Programming 2

PROJECT ASSIGNMENT 01 SYSTEM DESIGN & REPORT

GYM AND FITNESS MANAGEMENT SYSTEM

Project Information	Details
Semester	Summer 2025-2026 / 2026 Submission
Section	
Group No.	
Supervisor	
Prepared By	Project Group (member names and IDs to be inserted if required by instructor)

Submission package includes a GitHub-ready C# WinForms solution, SQL Server schema, diagrams, 17 UI mockups, README and this report.

CO2 / CO3 Rubric Mapping

Course Outcome / Criterion	Evidence in this Project
CO2 - Requirement Fulfilment	Three mandatory roles, domain-specific case study, all required marketplace features, matching navigation and database design.
CO2 - Validation	Required fields, email/password rules, rating 1-5 CHECK, stock/price/date constraints, two validation-state mockups.
CO2 - Verification	End-to-end flow from login -> role dashboard -> database-backed feature; SQL queries map to real tables and user stories.
CO3 - Organization	GitHub-ready structure with README, docs/diagrams, docs/screenshots and database/schema.sql.
CO3 - Database Representation	12 tables, PK/FK/UNIQUE/CHECK, junction tables, 3NF explanation, JOIN/GROUP BY/HAVING/aggregates.
CO3 - GUI	17 consistent WinForms-style mockups with visible titles, Back/Close/Logout paths, DataGridView lists and ComboBox filters.

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Chapter 1 - Introduction & Case Study

1.1 Project Overview

Gym and Fitness Management System is a desktop marketplace application designed in C# Windows Forms with SQL Server integration. The platform connects customers with multiple gym and fitness businesses while a platform-level Super Admin manages marketplace quality and earns commission. The design deliberately follows exactly three login roles: Super Admin, Admin/Gym Owner and Customer. Trainer information is supported as a managed gym entity, not as a fourth login role.

1.2 Case Study

People who want to join a gym or buy fitness-related services often have to visit individual gym pages, call different businesses, compare prices manually, and ask separately about membership packages, personal training, group classes, supplements and accessories. Small gym owners also face problems because their membership offers, fitness services and products are often managed through disconnected spreadsheets, social media messages and handwritten records. This makes searching, stock tracking, sales reporting and customer feedback difficult. The Gym and Fitness Management System addresses these problems by acting as a single marketplace where approved fitness businesses can publish their offerings and customers can compare and purchase them from one place.

The platform has three actors. The Super Admin owns and moderates the marketplace. New gym businesses are approved or suspended by the Super Admin, who also manages master categories, sees platform-wide sales and commission, identifies low-rated gyms and moderates abusive reviews. The Admin is the owner of one gym or fitness centre. Each Admin can manage only the data belonging to that gym, including its profile, trainers, membership packages, services, products, stock, discounts, reviews and sales report. This separation is important because one business owner must never be able to view or change another business's private operational data. The Customer is the end user. Customers create accounts, browse items from every active gym, search and filter choices, read details and reviews, add items to a cart, checkout, view invoices and order history, and submit ratings from one to five with written feedback.

The main database entities are Users, Gyms, Categories, Trainers, Items, Offers, Carts, CartItems, Orders, OrderItems, Reviews and Payments. A User with the Admin role owns a Gym, while each Gym can list many Items and Trainers. Categories classify Items, and Offers provide time-limited discounts. A Customer owns a Cart, and the Cart contains Items through the CartItems junction table. When checkout succeeds, an Order is created. Because one Order can contain many Items and the same Item can appear in many different Orders, OrderItems is the mandatory junction table that stores quantity, transaction price, discount and subtotal. Customers can write Reviews for Items, and a paid Order has one Payment record.

Money enters the system when a Customer pays for products, services or membership packages. The order stores the total amount and the platform commission. Each Gym has a commission rate, set to ten percent in the sample data. The Gym Owner's earnings report therefore shows the business's share after platform commission, while the Super Admin dashboard reports total paid orders, gross sales and commission across the marketplace. This model allows customers to discover fitness options more easily, gives gym owners a structured business-management system, and gives the platform owner a clear revenue and moderation mechanism.

1.3 Three Mandatory Roles

Role	Meaning in This Domain	Core Responsibility
Super Admin	Platform owner	Approvals, moderation, categories, platform sales and commission
Admin	Gym / fitness centre owner	Manage only own gym data, items,

		trainers, inventory, offers and earnings
Customer	Buyer / member	Browse all approved gyms, cart, checkout, invoice, rating and review

Chapter 2 - Functional Requirements & User Stories

2.1 Functional Requirements

Super Admin Functional Requirements

1. SA-FR01. The Super Admin shall log in using User ID/email and password and be routed to the Super Admin dashboard.
2. SA-FR02. The Super Admin shall view all registered gym businesses and their owners.
3. SA-FR03. The Super Admin shall approve pending gym businesses.
4. SA-FR04. The Super Admin shall suspend a gym business and its owner account.
5. SA-FR05. The Super Admin shall create and remove marketplace categories.
6. SA-FR06. The Super Admin shall view platform-wide paid orders, gross sales and commission.
7. SA-FR07. The Super Admin shall identify low-rated gyms using aggregated review data.
8. SA-FR08. The Super Admin shall moderate abusive/inappropriate reviews by hiding them.
9. SA-FR09. The Super Admin shall search/list platform data where relevant.
10. SA-FR10. The Super Admin shall update profile and password information.

Admin / Gym Owner Functional Requirements

11. AD-FR01. An Admin shall log in and be routed to the Gym Owner dashboard.
12. AD-FR02. An Admin shall only access data for the gym mapped to the logged-in owner account.
13. AD-FR03. An Admin shall update gym name, address and contact information.
14. AD-FR04. An Admin shall create, read, update and delete products, services and membership packages.
15. AD-FR05. An Admin shall manage trainers belonging to the gym.
16. AD-FR06. An Admin shall view stock and receive a low-stock list when `StockQty <= MinStock``.
17. AD-FR07. An Admin shall view sales details and net earnings after platform commission.
18. AD-FR08. An Admin shall create date-bounded percentage discounts/offers.
19. AD-FR09. An Admin shall view customer ratings and written reviews for its own gym.
20. AD-FR10. An Admin shall update personal profile/password information.

Customer Functional Requirements

21. CU-FR01. A Customer shall create an account with validated name, email and password.
22. CU-FR02. A Customer shall log in and be routed to the Customer dashboard.
23. CU-FR03. A Customer shall browse items from all approved gyms.
24. CU-FR04. A Customer shall search by gym name, item name or keyword.
25. CU-FR05. A Customer shall filter with dropdowns for category, item type and price range and may filter by location.
26. CU-FR06. A Customer shall view item details, price, offer and existing reviews.
27. CU-FR07. A Customer shall add items to a cart, change quantity by adding again, and remove cart lines.
28. CU-FR08. A Customer shall checkout using a payment method and create a paid order.
29. CU-FR09. Checkout shall reduce physical product stock while not reducing virtual service/membership availability.
30. CU-FR10. A Customer shall view order history and an invoice-style order detail.
31. CU-FR11. A Customer shall submit or update a 1-5 rating and written review.
32. CU-FR12. A Customer shall view active discounts automatically in browse/details/cart calculations.
33. CU-FR13. A Customer shall update profile/password information.

2.2 User Stories

Role-based login

As a registered user, I want to log in with my ID/email and password so that I am taken to the correct dashboard. The system hashes the entered password, validates it against `Users`, checks account status, reads `UserType`, and routes to Super Admin, Admin or Customer. Invalid or suspended accounts remain on the login screen with an error message.

Customer registration

As a new customer, I want to create an account so that I can buy fitness services and products. The form validates required fields, checks that the email looks valid and requires at least six password characters. A successful registration inserts a `Customer` record with active status.

Approve gym owner

As a Super Admin, I want to approve a pending gym so that a legitimate business can sell on the platform. I open Manage Gym Owners, select a pending gym, press Approve and the gym becomes Active. The owner account remains usable only when its status is Active.

Suspend gym owner

As a Super Admin, I want to suspend a gym and its owner so that policy violations can be controlled. Selecting a business and pressing Suspend updates the gym and the linked owner account. Suspended owners cannot log in and suspended gyms disappear from customer browsing.

Manage categories

As a Super Admin, I want to manage categories so that all gyms use a consistent classification. I can add a unique category name and description. Deletion is blocked by the database when a category is still referenced by an item.

Platform commission report

As a Super Admin, I want to see gross sales and platform commission so that I can measure marketplace performance. The report aggregates paid orders with `SUM`, `COUNT` and joins. Commission is stored per order and gym rates remain visible in the business table.

Low-rated gyms

As a Super Admin, I want to find low-rated gyms so that quality issues can be investigated. Review scores are grouped by gym and an SQL `HAVING` condition identifies averages below the threshold. Only visible reviews contribute to the result.

Review moderation

As a Super Admin, I want to hide abusive reviews so that the public marketplace remains appropriate. The review remains in the database for audit history, but `IsVisible` is set to false. Gym owners can only view reviews and cannot perform moderation.

Gym profile

As a Gym Owner, I want to edit my gym profile so that customers receive accurate information. The owner can update gym name, address and contact information. Every update targets the logged-in `GymId`, preventing access to another gym.

Item CRUD

As a Gym Owner, I want to manage products, services and memberships so that my current offerings appear in the marketplace. I choose a category and type, provide price, stock thresholds and description, and save. The grid refreshes after create/update/delete operations.

Trainer management

As a Gym Owner, I want to maintain trainer information so that customers can understand the expertise available at my gym. Trainers have name, specialization, phone, experience and availability. Trainer is a managed entity rather than a fourth system login role, preserving the required exactly-three-role model.

Low stock

As a Gym Owner, I want to see products at or below minimum stock so that I can restock before sales are lost. The inventory dashboard filters physical products with ``StockQty <= MinStock``. Services and memberships are not treated as physical stock.

Gym sales report

As a Gym Owner, I want to see my own sales and net earnings so that I can evaluate the business. ``OrderItems`` is filtered by the owner gym's ``GymId``. The report shows customer, item, quantity, unit price and subtotal, while net earnings subtract platform commission.

Create offer

As a Gym Owner, I want to create a timed discount so that I can attract customers. I select one of my items, enter a percentage and start/end dates. Validation prevents invalid percentages and end dates before start dates.

View reviews

As a Gym Owner, I want to read customer feedback so that I can improve service quality. The review list is restricted to items belonging to the current gym. The owner has no delete/hide button, separating business feedback from platform moderation.

Browse marketplace

As a Customer, I want to browse all active gyms so that I can compare choices. The browse query only returns available items from active gyms. Gym name, category, price, stock/location, rating and current discount are displayed in a ``DataGridView``.

Search

As a Customer, I want to search by gym, item or keyword so that I can quickly find relevant fitness options. The application searches item name, gym name and description using a parameterized query. Clearing the search displays the full marketplace again.

Multi-filter browsing

As a Customer, I want dropdown filters so that I can narrow choices without typing complex queries. Category, item type and price range are ComboBoxes, with location as an additional text filter. All selected conditions are applied together in the SQL query.

Item details

As a Customer, I want to open an item and read details/reviews so that I can make an informed decision. The details screen shows gym, category, description, original price, active discount and final price. Existing visible customer reviews are shown below.

Add to cart

As a Customer, I want to add an item and quantity to my cart so that I can purchase multiple choices together. The system creates one cart per customer if needed. Adding the same item again increases its quantity instead of creating a duplicate cart line.

Remove from cart

As a Customer, I want to remove a selected cart line so that I can correct my purchase before payment. The selected `CartItemId` is deleted and the total is recalculated. The cart remains linked to only the logged-in customer.

Checkout

As a Customer, I want to pay for my cart so that an order is created. Checkout runs inside a SQL transaction, validates physical stock, calculates discounts and platform commission, inserts `Orders`, `OrderItems` and `Payments`, reduces product stock, clears the cart, and commits only when all operations succeed.

Order history and invoice

As a Customer, I want to view previous orders and invoice details so that I have a purchase record. The order history is filtered by the current customer. Selecting an order shows each item, gym, quantity and subtotal and calculates the total invoice amount.

Rating and review

As a Customer, I want to rate an item from 1 to 5 and write a comment so that I can share my experience. The database has a `CHECK` constraint for the rating range and a unique customer-item rule. Submitting again updates the existing review rather than duplicating it.

Profile and password

As any logged-in user, I want to update my profile and optionally change my password so that my account information remains current. Name is required and a new password must contain at least six characters. Passwords are stored as SHA-256 hashes rather than plain text.

Chapter 3 - UI Navigation Diagram

The application starts at sign up/login. After successful credential validation, a diamond decision node checks `UserType` and routes to exactly one of three dashboards. Every child screen has a Close/Back path and each dashboard has Logout -> Login. Customer purchase flow continues from browse to details/cart, checkout, payment and order history/invoice. Validation failures return to the relevant form without changing data.

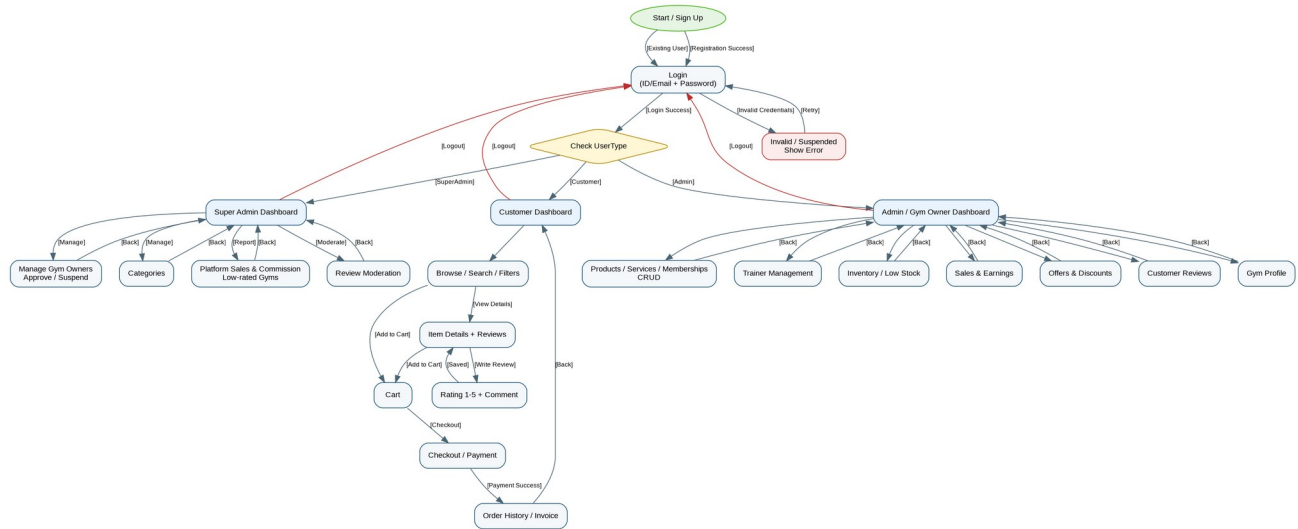


Figure 3.1 - Complete UI navigation with role-decision branch and labelled arrows.

Chapter 4 - Database Design

4.1 SQL Schema Diagram

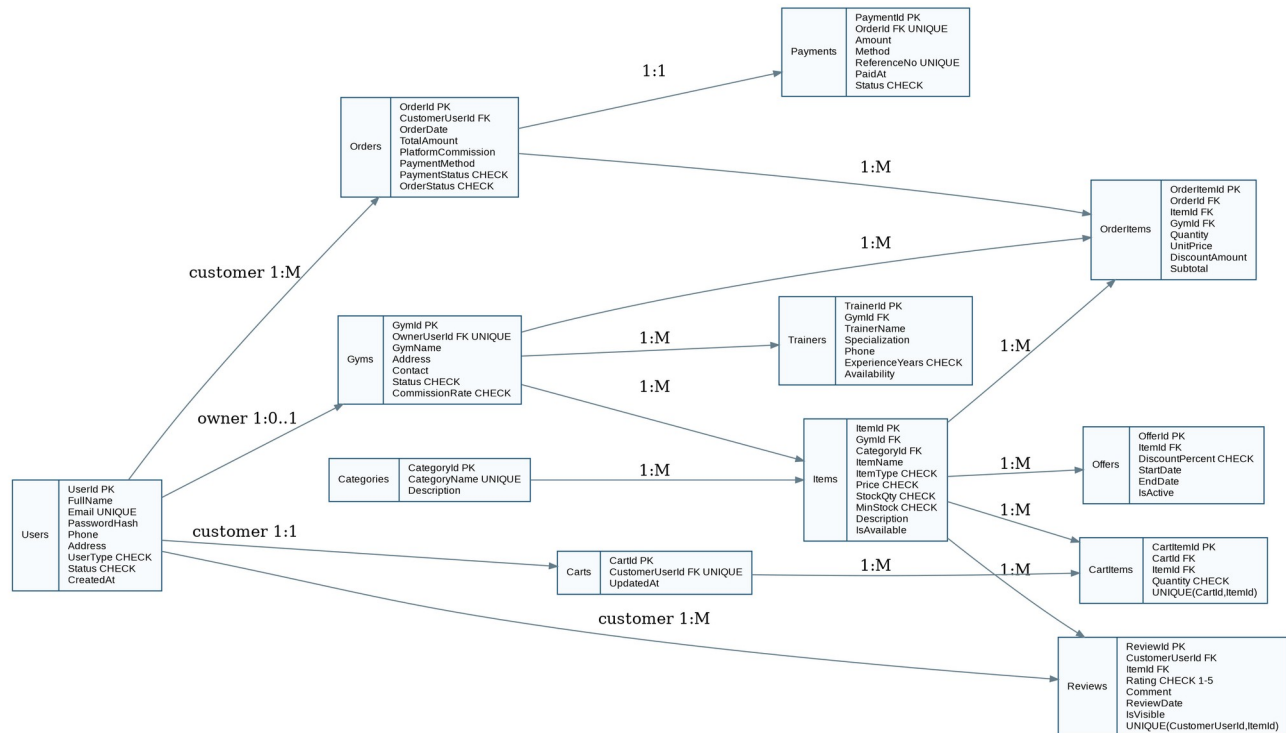


Figure 4.1 - SQL schema diagram showing PK/FK relationships and major constraints.

4.2 Table Descriptions

Table	Purpose	Important Constraints
Users	Stores all three login roles	UserId PK, Email UNIQUE, UserType CHECK, Status CHECK
Gyms	One business record per Admin owner	GymId PK, OwnerUserId FK UNIQUE, commission/status checks
Categories	Master item categories	CategoryId PK, CategoryName UNIQUE
Trainers	Gym trainer profiles	GymId FK, ExperienceYears CHECK
Items	Products/services/memberships	GymId FK, CategoryId FK, ItemType/price/stock checks
Offers	Date-bounded discounts	ItemId FK, discount/date checks
Carts	One active cart per Customer	CustomerUserId FK UNIQUE
CartItems	Cart-to-item junction	CartId FK, ItemId FK, UNIQUE pair, Quantity CHECK
Orders	Completed purchase header	CustomerUserId FK, total/commission/payment/order status
OrderItems	Mandatory order-to-item junction	OrderId/ItemId/GymId FKs; qty, price, discount, subtotal
Reviews	Customer ratings/comments	Rating CHECK 1-5, unique Customer+Item
Payments	Payment record	OrderId FK UNIQUE, amount/status/reference constraints

4.3 ER Diagram (Bonus Quality)

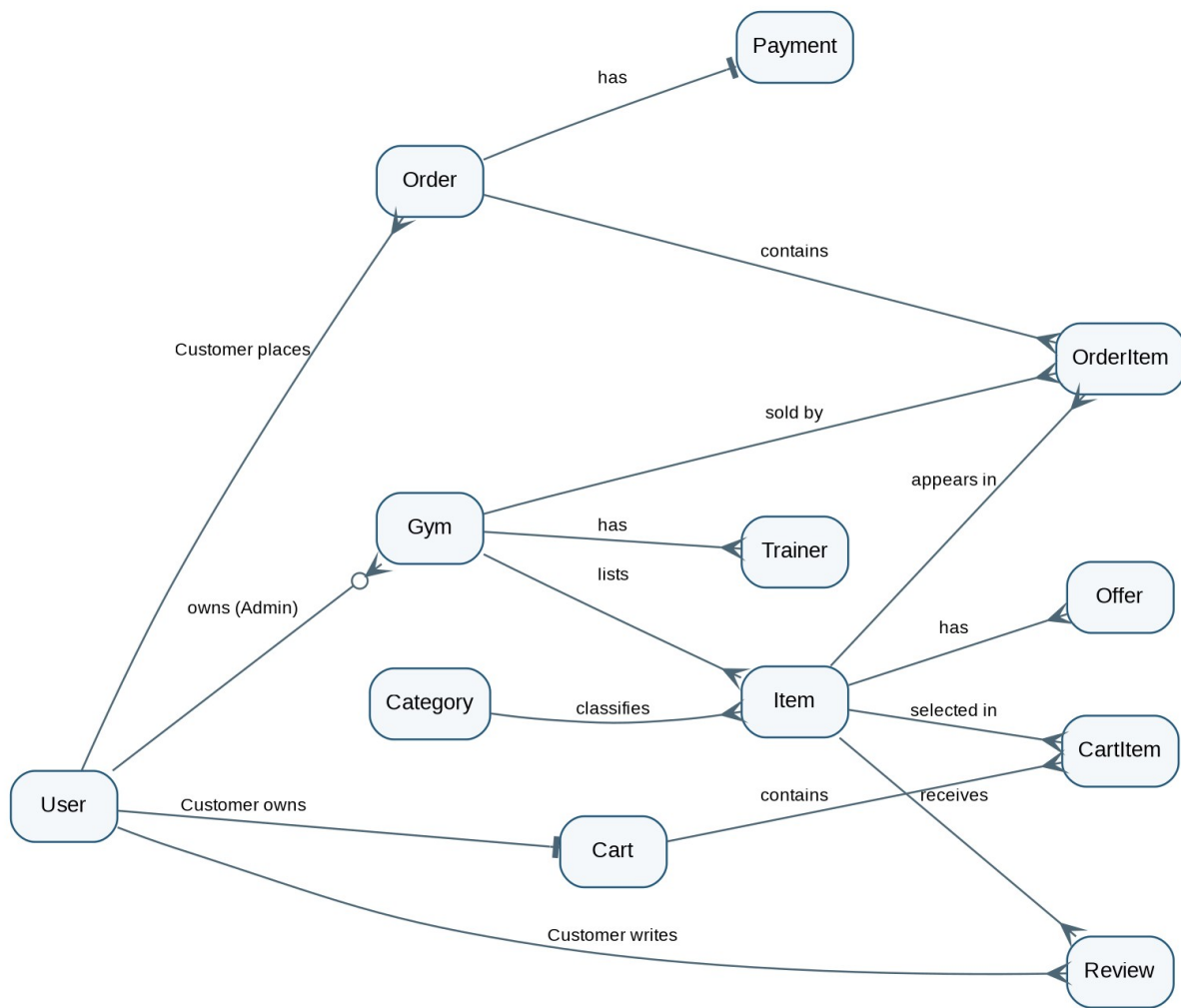


Figure 4.2 - Entity-relationship view of the marketplace model.

4.4 Normalisation to 3NF

First Normal Form: Every field stores one atomic value. Repeating lists such as cart items and order items are separated into child/junction tables.

Second Normal Form: Non-key attributes depend on the whole key. For example, transaction attributes such as `Quantity`, `UnitPrice`, and `Subtotal` belong to `OrderItems`, not to `Items` or `Orders` alone.

Third Normal Form: Transitive facts are separated. `GymName` is stored only in `Gyms` and referenced by `GymId`; `CategoryName` is stored only in `Categories`; owner details are stored in `Users`. This avoids update anomalies and duplicated descriptive data.

4.5 Data Isolation

Every Admin operational query is filtered by the logged-in `GymId`. Items, trainers, offers, stock, reviews and sales therefore return only records belonging to that owner. This prevents a Gym Owner from reading or modifying another Gym Owner's private data.

Chapter 5 - SQL Script & Feature Queries

The complete runnable file is `database/schema.sql`. It creates the database, tables, constraints, seed data and demonstration queries. The implementation uses parameterized ADO.NET queries in C# to avoid concatenating user input into SQL.

5.1 Login verification

```
SELECT u.UserId,u.FullName,u.UserType,g.GymId
FROM Users u LEFT JOIN Gyms g ON g.OwnerUserId=u.UserId
WHERE (u.Email=@id OR CAST(u.UserId AS varchar(20))=@id)
AND u.PasswordHash=@pw AND u.Status='Active';
```

This query directly supports the named system feature and references tables that appear in the schema diagram.

5.2 Search / Filter

```
SELECT i.ItemId,g.GymName,i.ItemName,i.ItemType,c.CategoryName,i.Price
FROM Items i JOIN Gyms g ON g.GymId=i.GymId
JOIN Categories c ON c.CategoryId=i.CategoryId
WHERE g.Status='Active' AND i.IsAvailable=1
AND (@q='' OR i.ItemName LIKE '%'+@q+'%' OR g.GymName LIKE '%'+@q+'%');
```

This query directly supports the named system feature and references tables that appear in the schema diagram.

5.3 Gym earnings - JOIN + GROUP BY + SUM

```
SELECT g.GymName, SUM(oi.Subtotal) GrossSales,
SUM(oi.Subtotal*(1-g.CommissionRate)) GymEarnings
FROM Gyms g JOIN OrderItems oi ON oi.GymId=g.GymId
JOIN Orders o ON o.OrderId=oi.OrderId AND o.PaymentStatus='Paid'
GROUP BY g.GymName,g.CommissionRate;
```

This query directly supports the named system feature and references tables that appear in the schema diagram.

5.4 Low-rated gyms - GROUP BY + HAVING + AVG

```
SELECT g.GymName,AVG(CAST(r.Rating AS decimal(5,2))) AverageRating
FROM Gyms g JOIN Items i ON i.GymId=g.GymId
JOIN Reviews r ON r.ItemId=i.ItemId AND r.IsVisible=1
GROUP BY g.GymName
HAVING AVG(CAST(r.Rating AS decimal(5,2))) < 3.5;
```

This query directly supports the named system feature and references tables that appear in the schema diagram.

5.5 Low stock alert

```
SELECT g.GymName,i.ItemName,i.StockQty,i.MinStock
FROM Items i JOIN Gyms g ON g.GymId=i.GymId
WHERE i.ItemType='Product' AND i.StockQty<=i.MinStock;
```

This query directly supports the named system feature and references tables that appear in the schema diagram.

5.6 Active offers

```
SELECT g.GymName,i.ItemName,o.DiscountPercent,o.StartDate,o.EndDate
FROM Offers o JOIN Items i ON i.ItemId=o.ItemId
```

```
JOIN Gyms g ON g.GymId=i.GymId  
WHERE o.IsActive=1 AND SYSDATETIME() BETWEEN o.StartDate AND o.EndDate;
```

This query directly supports the named system feature and references tables that appear in the schema diagram.

Chapter 6 - User Interface Design

All mockups use one consistent blue/white/green visual language, visible screen titles, standardized buttons and grid layouts. Data lists use DataGridView-style tables and the customer filters use ComboBox-style dropdowns. Validation/error states are explicitly shown in the Sign Up and Checkout mockups.

Login

Gym and Fitness Management System

Sign in to your account

User ID or Email

admin@gym.com

Password

Login

Create Account

Demo: admin@gym.com / admin123

Login form - entry point for all three roles.

Customer Sign Up

Gym and Fitness Management System

Full Name

A Customer

Email

customer.example.com

Invalid email address. Use a valid email such as name@example.com

Password

Password must be at least 6 characters.

Phone

01700000000

Address

Dhaka

Create Account

Back

Customer sign-up form showing email and password validation states.

Super Admin Dashboard

Gym and Fitness Management System

Paid Orders

128

Gross Sales

Tk 842,500

Platform Commission

Tk 84,250

Selling Gyms

12

Manage Gym Owners

Categories

Platform Reports

Review Moderation

Profile

Logout

Platform controls approval, marketplace categories, commission reporting and review moderation.

Super Admin dashboard with platform summary and control actions.

Manage Gym Owners

Gym and Fitness Management System

Refresh

Approve

Suspend

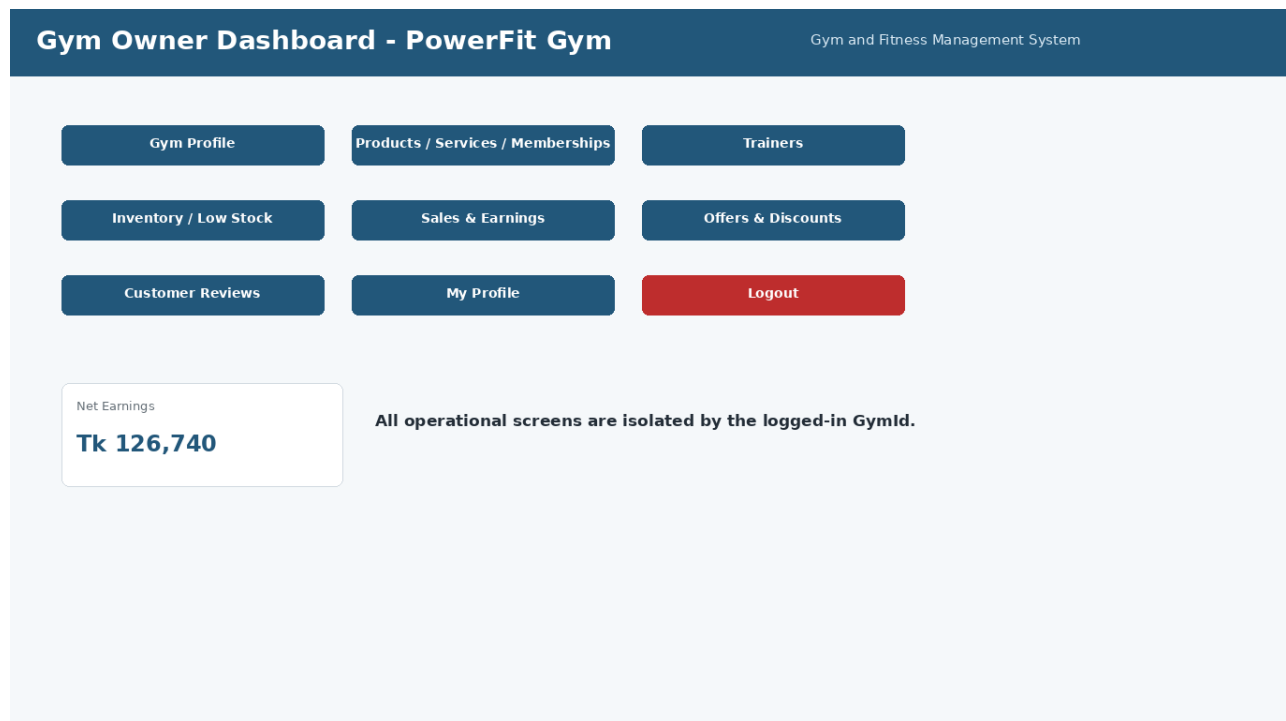
Close

GymId	Gym Name	Owner	Email	Address	Status	Commission
1	PowerFit Gym	PowerFit Owner	owner1@gym.com	Dhanmondi	Active	10%
2	Iron Paradise	Iron Paradise Owner	owner2@gym.com	Banani	Active	10%
3	Fit Nation	New Owner	newowner@gym.com	Uttara	Pending	10%

Super Admin - approve/suspend gym-owner businesses.



Platform sales, commission and low-rated gym report.



Admin/Gym Owner dashboard with data-isolated business modules.

Product, Service & Membership CRUD

Gym and Fitness Management System

Name

Category

Type

Price

Stock

Min

Whey Protein 1kg

Supplements ▼

Product ▼

4200

4

5

Add

Update

Delete

Close

ID	Item	Type	Category	Price	Stock	Min Stock	Available
11	Monthly Membership	Membership	Membership	2500	9999	0	Yes
12	Personal Training Session	Service	Personal Training	1200	9999	0	Yes
13	Whey Protein 1kg	Product	Supplements	4200	4	5	Yes

Admin product/service/membership CRUD using dropdowns and a data grid.

Trainer Management

Gym and Fitness Management System

Trainer Name

Specialization

Phone

Experience

Availability

Rahim Khan

Strength & Conditioning

01910000001

5

Sun-Thu 5PM-10PM

Add Trainer

Delete

ID	Trainer	Specialization	Phone	Experience	Availability
1	Rahim Khan	Strength & Conditioning	01910000001	5 years	Sun-Thu 5PM-10PM
2	Nabila Islam	Yoga & Mobility	01910000002	4 years	Sat-Wed 7AM-12PM

Trainer is a managed entity; it is not a fourth login role.

Admin trainer management. Trainer is an entity, not a fourth login role.

Inventory Dashboard - Low Stock Alert

Gym and Fitness Management System

Products at or below minimum stock

Refresh

Close

Item ID	Product	Type	Stock Qty	Min Stock	Alert
13	Whey Protein 1kg	Product	4	5	LOW STOCK
21	Resistance Band Set	Product	2	3	LOW STOCK

Inventory dashboard showing low-stock alert condition.

Sales & Earnings Report

Gym and Fitness Management System

Net Earnings after Commission

Tk 126,740

Refresh

Close

Order	Date	Customer	Item	Qty	Unit Price	Subtotal
1007	06 Sep 2026	Customer One	Monthly Membership	1	Tk 2,500	Tk 2,250
1006	05 Sep 2026	Customer Two	Whey Protein 1kg	2	Tk 4,200	Tk 8,400
1005	05 Sep 2026	Customer Three	PT Session	3	Tk 1,200	Tk 3,600

Gym-specific sales and net earnings report.

Browse, Search & Filter All Gyms

Gym and Fitness Management System

Search

membership

Category

Membership

Item Type

All Types

Price Range

1000 - 3000

Location

Dhaka

Search / Filter

View Details

ID	Gym	Item	Type	Category	Price	Rating	Discount
11	PowerFit Gym	Monthly Membership	Membership	Membership	Tk 2,500	4.8	10%
19	Iron Paradise	Monthly Membership Plus	Membership	Membership	Tk 3,000	4.4	0%

Add Selected to Cart

Close

Customer marketplace browse/search with category, type, price and location filters.

Item Details & Reviews

Gym and Fitness Management System

Monthly Membership

PowerFit Gym | Membership | Dhanmondi, Dhaka

Price: Tk 2,500 **Offer: 10%** **Final: Tk 2,250**

Full gym access for one month.

Quantity

1

Add to Cart

Write Review

Customer Reviews

Customer	Rating	Comment	Date
Customer One	5 / 5	Clean gym, useful equipment and friendly trainers.	05 Sep 2026
Customer Two	4 / 5	Good location and helpful staff.	03 Sep 2026

Customer item details with current offer and public reviews.

Shopping Cart

Gym and Fitness Management System

Cart Total

Tk 10,650

Remove Selected

Checkout

Gym	Item	Qty	Price	Discount	Line Total
PowerFit Gym	Monthly Membership	1	Tk 2,500	10%	Tk 2,250
PowerFit Gym	Whey Protein 1kg	2	Tk 4,200	0%	Tk 8,400

Customer cart with calculated line totals and checkout action.

Checkout & Payment

Gym and Fitness Management System

Amount to Pay: Tk 10,650

Payment Method

Mobile Banking

Reference / Account

Payment reference is required for Mobile Banking.

Confirm Payment

Cancel

Checkout/payment mockup showing a validation/error state.

Order History & Invoice

Gym and Fitness Management System

Refresh

View Invoice

Close

Order	Date	Total	Method	Payment	Status
1007	06 Sep 2026	Tk 10,650	Mobile Banking	Paid	Confirmed
1001	28 Aug 2026	Tk 2,250	Card	Paid	Completed

INVOICE - ORDER #1007

Customer: Customer One

Monthly Membership x1 Tk 2,250

Whey Protein 1kg x2 Tk 8,400

TOTAL: Tk 10,650

Order history with printable invoice-style detail.

Rate & Review

Gym and Fitness Management System

Monthly Membership - PowerFit Gym

Rating

5

Comment

Clean gym, useful equipment and friendly trainers.

Submit Review

Cancel

Customer 1-5 rating and written review form.

Offers & Discounts

Gym and Fitness Management System

Item

Discount %

Start Date

End Date

Create Offer

Delete

Close

Monthly Membership

10

01/09/2026

30/09/2026

Offer ID	Item	Discount	Start Date	End Date	Active
1	Monthly Membership	10%	01 Sep 2026	30 Sep 2026	Yes
2	Whey Protein 1kg	5%	05 Sep 2026	15 Sep 2026	Yes

Admin date-bounded percentage offers/discounts management.

Chapter 7 - Work Distribution

The assignment requires a truthful contribution table and separate GitHub commits for all group members. Personal group details were not supplied in the chat, so the technical contribution plan is shown below without inventing student identities.

Member	Student ID	Suggested / Actual Contribution
Member 1	_____	Case study, functional requirements, Super Admin UI and report integration
Member 2	_____	Database schema, SQL queries, ER/schema diagrams and data validation
Member 3	_____	Admin/Customer UI mockups, navigation diagram and README/GitHub structure
Member 4 (if applicable)	_____	Testing, consistency review, screenshots and final PDF proofread

Important: GitHub should contain genuine commits from each group member. Do not fabricate contribution history. The repository structure is already prepared so files can be committed by the appropriate contributors.

Chapter 8 - Conclusion & Future Work

The Gym and Fitness Management System design satisfies the required three-role marketplace model and aligns the case study, navigation, database, SQL queries and UI mockups around the same data flow. Super Admin controls the platform, Admin owners manage only their own gym, and Customers complete the full discovery-to-purchase-to-review journey. The database uses normalized tables, clear PK/FK relationships, the mandatory OrderItems junction table, business constraints and SQL aggregates required for reporting. The accompanying C# WinForms source also implements role routing and the main database-backed workflows as bonus-quality preparation for the later coding phase.

8.1 Future Work

- Add real online payment gateway integration instead of payment simulation.
- Add class/session scheduling and capacity reservations as an extension to the existing service item model.
- Add trainer assignment/booking history if the later coding assignment explicitly requires it.
- Generate PDF invoices directly from the application.
- Add dashboard charts, audit logging, password salting and role/permission hardening for production use.
- Deploy the SQL database to a hosted environment and add automated backup/restore.

8.2 Expected Challenges

The main implementation challenges are keeping gym-owner data isolated, keeping cart/order/stock changes transaction-safe, calculating discounts and commission consistently, validating all user input, and maintaining agreement between UI controls, database columns and SQL queries. The supplied design addresses these challenges by using GymId filters, SQL transactions, constraints, parameterized queries and one shared schema.

Appendix A - Demo Accounts

Role	Email	Password
Super Admin	admin@gym.com	admin123
Admin / Gym Owner	owner1@gym.com	owner123
Admin / Gym Owner	owner2@gym.com	owner123
Customer	customer1@gym.com	cust123

Appendix B - Repository Structure

```
Gym_and_Fitness_Management_System/  
├── GymAndFitnessManagementSystem.sln  
├── GymAndFitnessManagementSystem/  
│   ├── Forms/  
│   ├── Data/  
│   ├── Models/  
│   ├── Common/  
│   └── App.config  
├── database/schema.sql  
├── docs/  
│   ├── Project_Report.pdf  
│   ├── diagrams/  
│   └── screenshots/  
└── README.md
```

Appendix C - Requirement Checklist

- Project domain: Gym & Fitness
- Exactly 3 login roles: Super Admin, Admin/Gym Owner, Customer
- Multi-gym marketplace concept
- Case study included in final report
- Functional requirements included
- User stories included
- UI navigation diagram included
- SQL schema diagram included
- ER diagram included as bonus-quality material
- Database is 3NF and has more than 6 tables
- PK, FK, UNIQUE and CHECK constraints used
- `OrderItems` junction table included
- SQL file includes sample data: 1 Super Admin, 2 Admins, 3 Customers
- SQL examples include JOIN, GROUP BY, HAVING, SUM, AVG, COUNT
- Search and at least 3 filters included
- Product/service/membership CRUD included
- Cart, checkout, invoice/order history included
- Rating 1-5 + written review included
- Offers/discounts included
- Low stock alert included
- Platform commission included
- Data isolation by GymId included
- 12+ UI mockups/screenshots included
- Validation states included in UI mockups/report

- GitHub-ready repository structure included
- Working role-routing C# source included as bonus implementation